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(54) **INTEGRATED CIRCUIT CHIP COMPRISING
A RADIOFREQUENCY COMPONENT**

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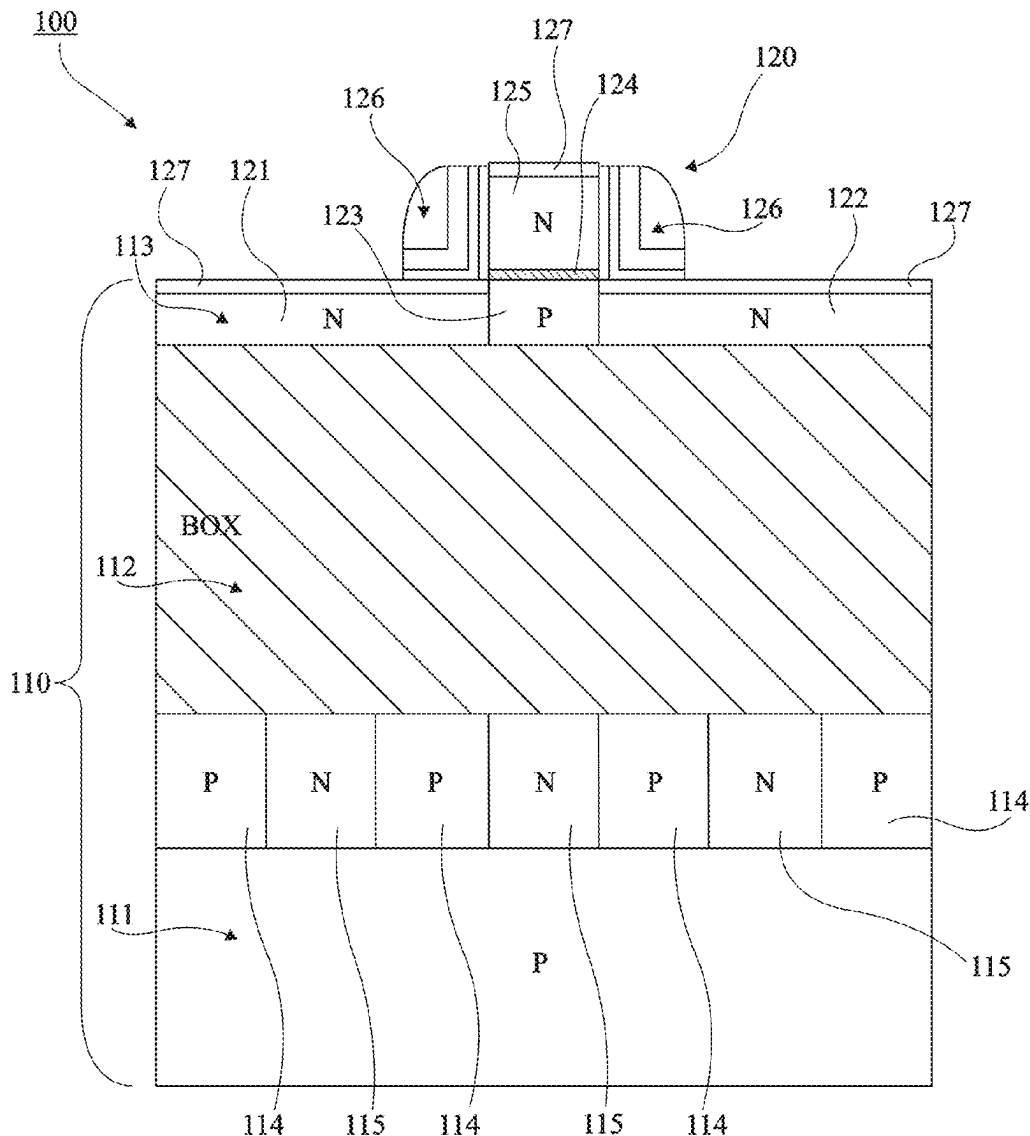
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(57)

ABSTRACT

The present description concerns an integrated circuit chip comprising at least one component, arranged inside and/or on top of a structure comprising a semiconductor substrate on which rests an insulating layer having a semiconductor layer resting thereon, wherein at least two PN junctions are arranged at the interface between the substrate and the insulating layer.



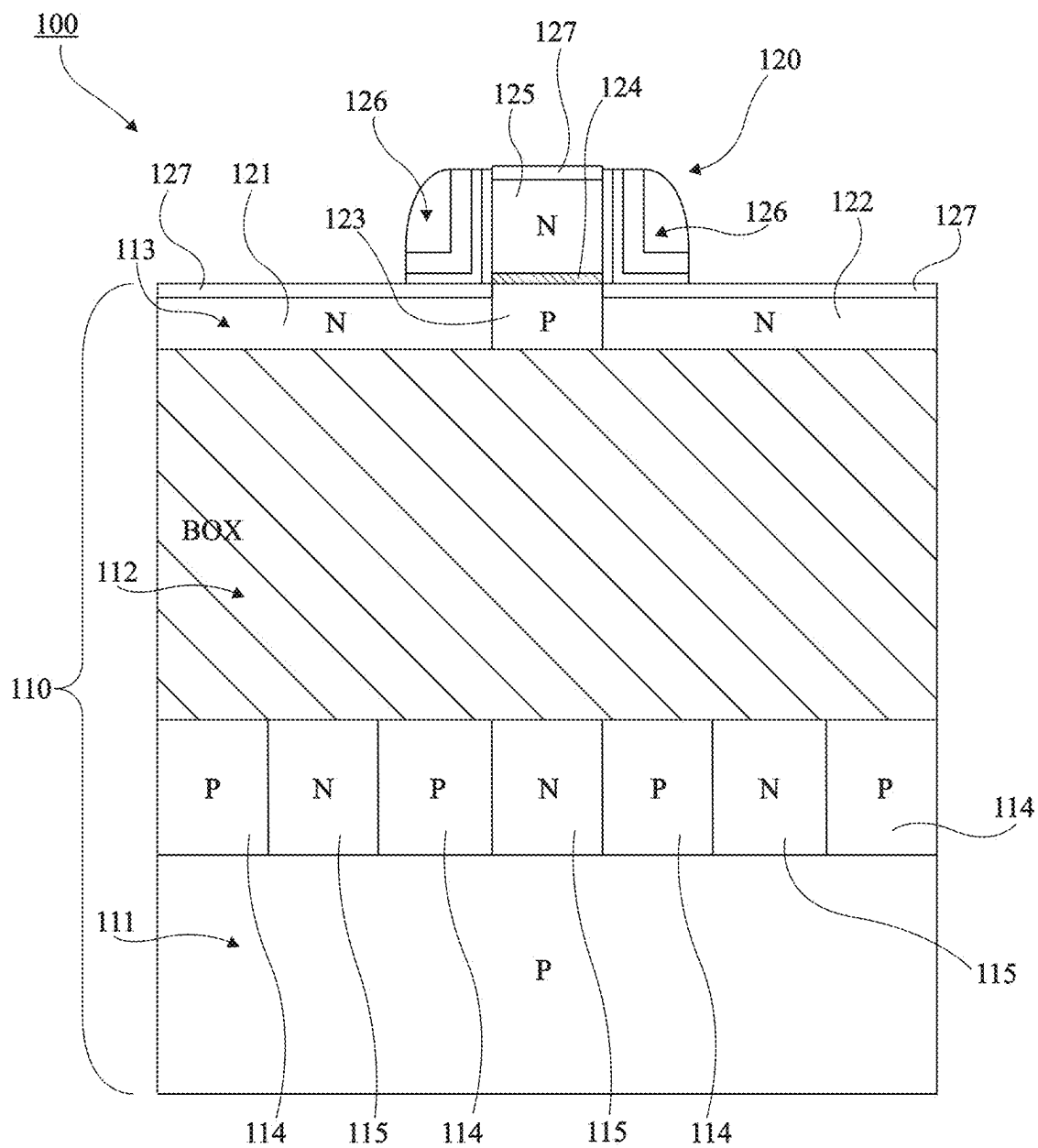


Fig 1

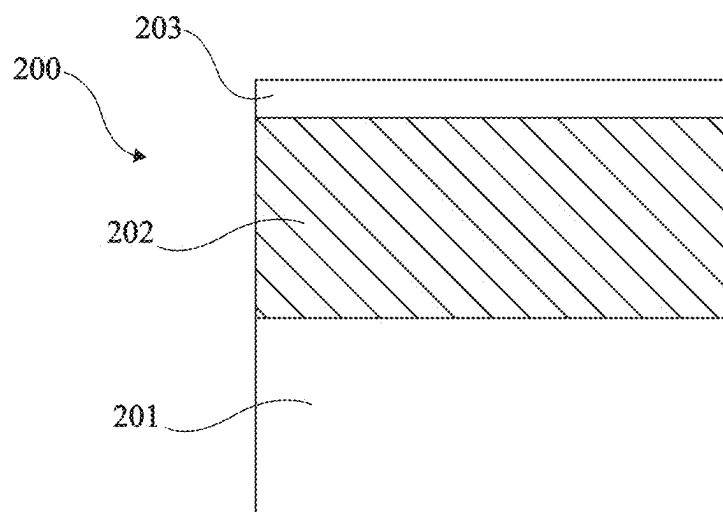


Fig 2

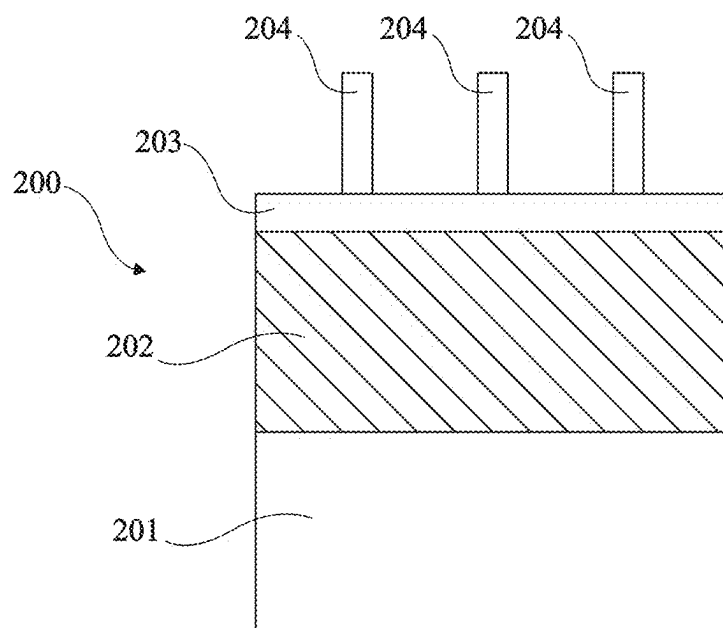


Fig 3

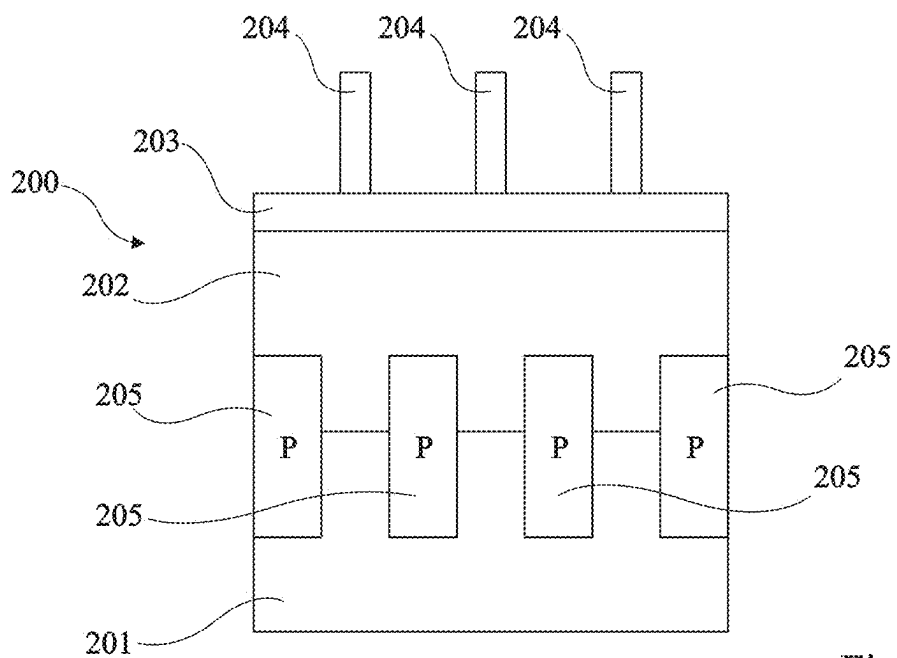


Fig 4

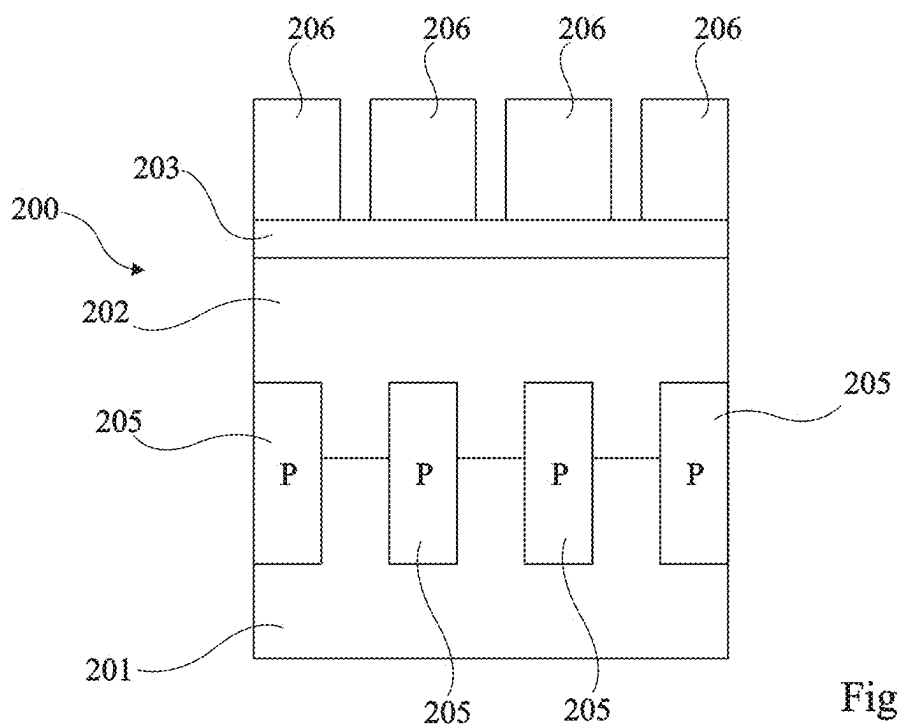


Fig 5

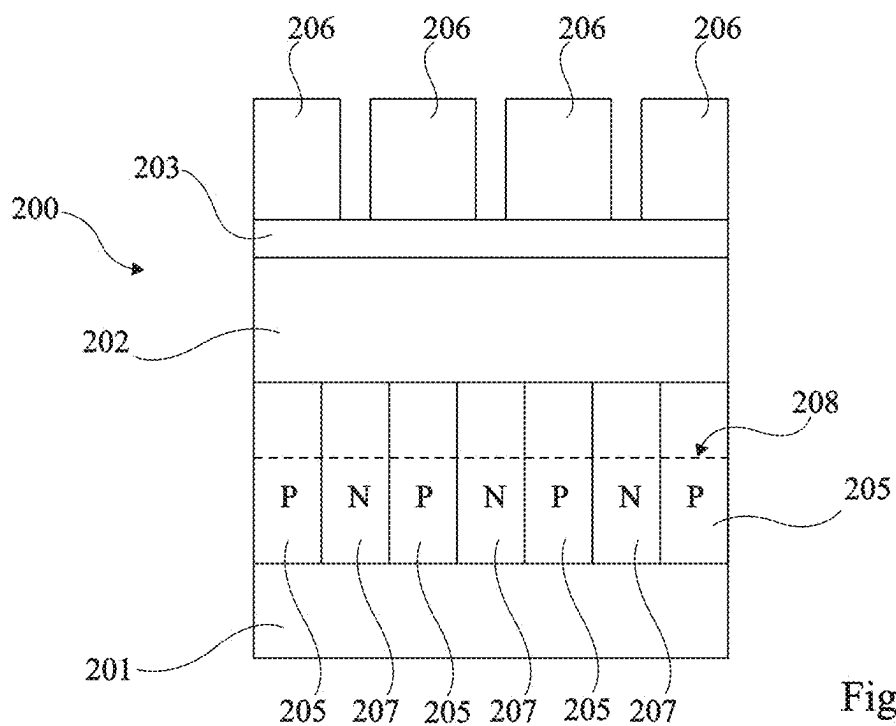


Fig 6

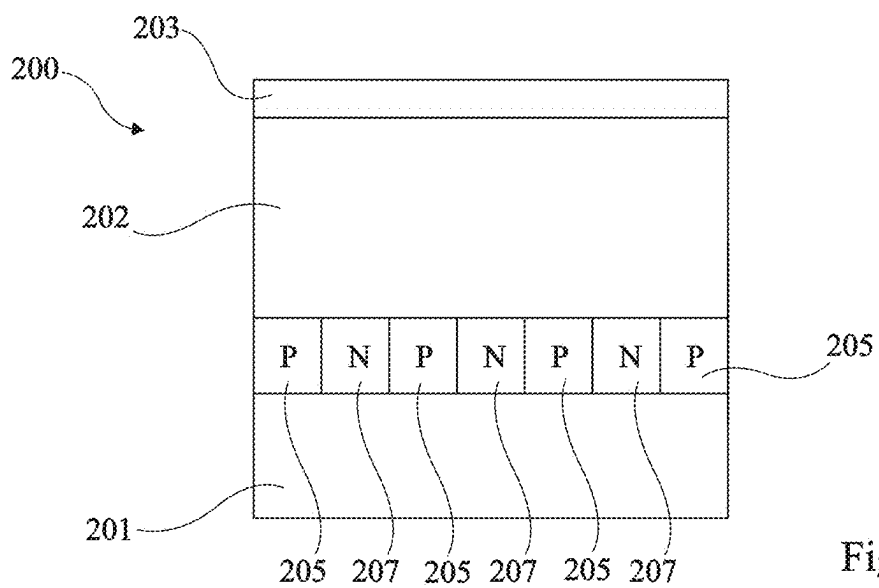


Fig 7

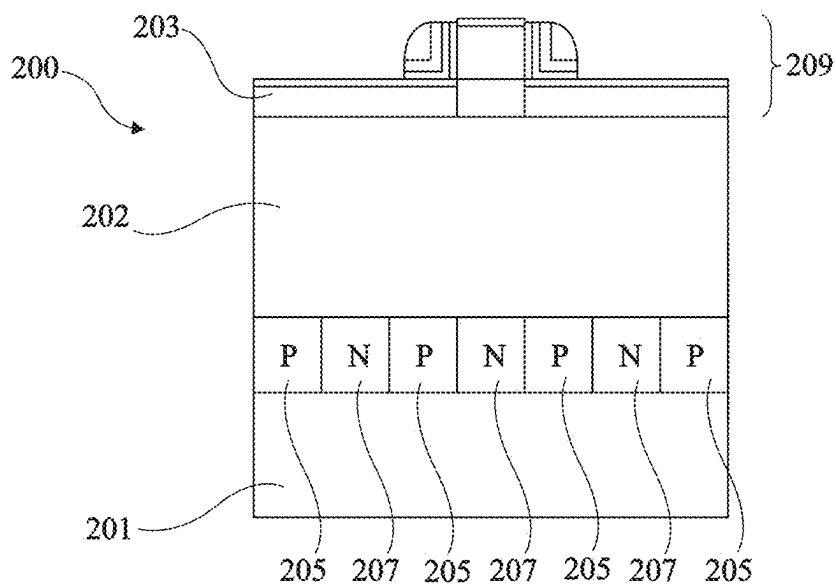


Fig 8

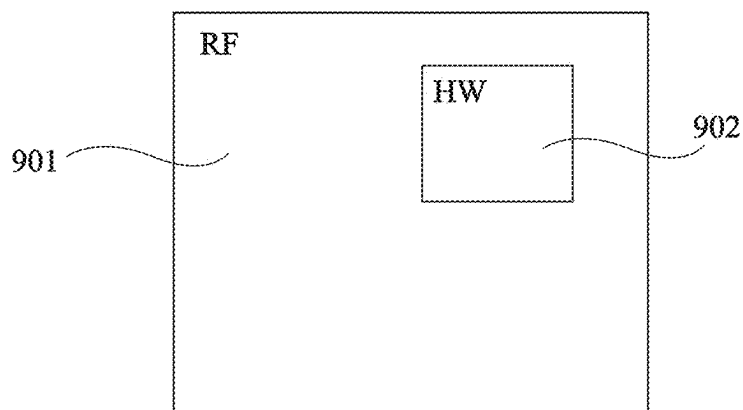


Fig 9

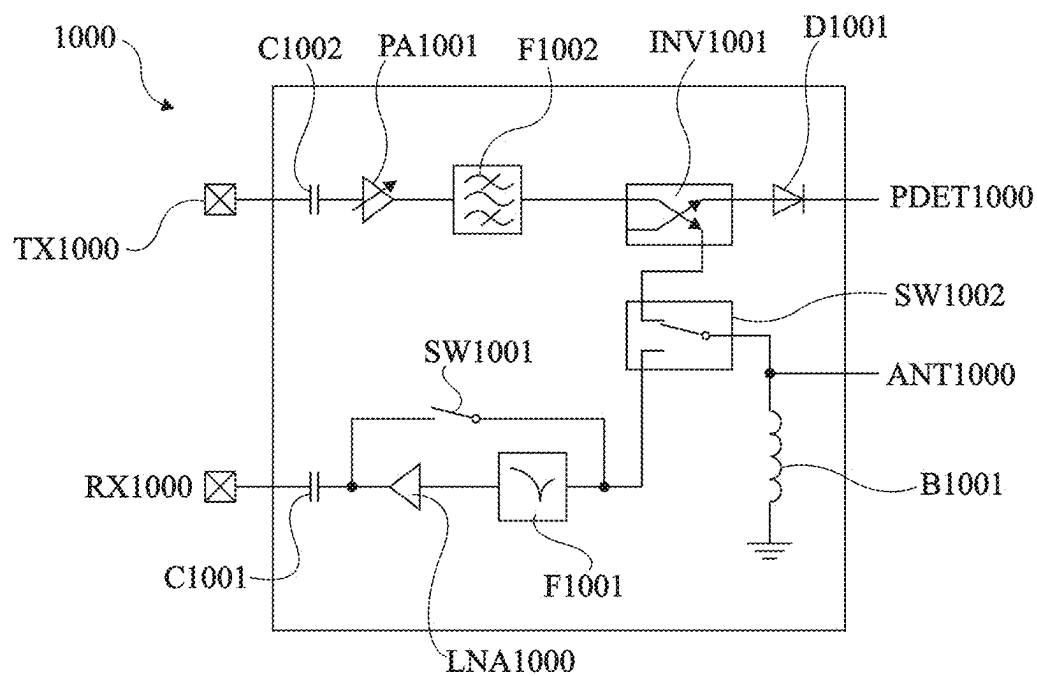


Fig 10

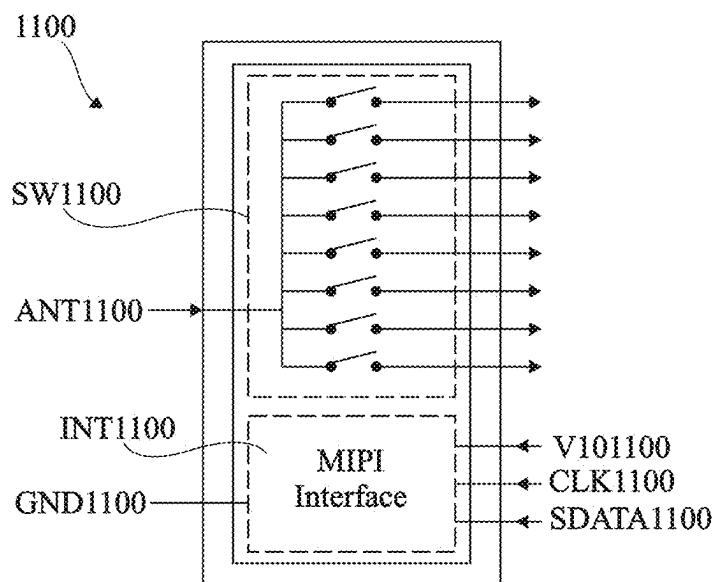


Fig 11

INTEGRATED CIRCUIT CHIP COMPRISING A RADIOFREQUENCY COMPONENT

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the priority benefit of French Patent Application No. 2402317, filed on Mar. 7, 2024, entitled “Circuit intégré comprend un composant radiofréquence,” which is hereby incorporated herein by reference to the maximum extent allowable by law.

TECHNICAL FIELD

[0002] The present disclosure generally concerns the field of integrated circuit chips and their manufacturing processes. It more specifically aims at the field of integrated circuit chips comprising a radio frequency component and their manufacturing methods.

BACKGROUND

[0003] One generally calls radio frequency components as components intended to use AC electrical signals at frequencies in the range from 3 kHz to 30 GHz. Among known radio frequency components, active components such as amplifiers, mixers, and data converters, and passive components such as capacitors, inductors, and antennas can be found.

[0004] It would be desirable to be able to improve, at least partly, certain aspects of integrated circuit chips comprising a radio frequency component.

SUMMARY

[0005] There exists a need for electronic components formed on a better electrically insulated SOI-type structure.

[0006] There exists a need for integrated circuit chips intended for radio frequency applications comprising such electronic components.

[0007] There exists a need for methods of manufacturing such electronic components.

[0008] An embodiment overcomes all or part of the disadvantages of known electronic components.

[0009] An embodiment overcomes all or part of the disadvantages of known integrated circuit chips intended for radio frequency applications.

[0010] An embodiment overcomes all or part of the disadvantages of known methods of manufacturing such chips.

[0011] An embodiment provides an electronic component formed inside and on top of an SOI-type structure in which PN junctions are formed at the interface between the support substrate and the buried oxide layer.

[0012] An embodiment provides an integrated circuit chip comprising such an electronic chip.

[0013] An embodiment provides a method of manufacturing such an electronic component.

[0014] An embodiment provides an integrated circuit chip comprising at least one component, arranged inside and/or on top of a structure comprising a semiconductor substrate on which rests an insulating layer having a semiconductor layer resting thereon, wherein at least two PN junctions are arranged at the interface between the substrate and the insulating layer.

[0015] According to an embodiment, the at least two PN junctions are arranged against the interface between the substrate and the insulating layer.

[0016] According to an embodiment, the PN junctions comprise an alternation of N-type doped wells and of P-type doped wells.

[0017] According to an embodiment, the N-type doped wells have a width greater than or equal to 78 nm.

[0018] According to an embodiment, the P-type doped wells have a width greater than or equal to 200 nm.

[0019] According to an embodiment, the semiconductor substrate has a resistivity greater than 125 ohm-cm.

[0020] According to an embodiment, the component is a switch.

[0021] According to an embodiment, the component is a MOS-type transistor.

[0022] According to an embodiment, the chip is adapted to radio frequency applications, the component being intended to use AC electrical signals at frequencies in the range from 3 kHz to 30 GHz.

[0023] According to an embodiment, the chip is an SPDT-type switch, the component being a switch.

[0024] According to an embodiment, the chip is a radio frequency switch with a frequency band selection, the component being a switch.

[0025] Another embodiment provides a method of manufacturing an integrated circuit chip comprising at least one component, and being formed inside and/or on top of a structure comprising a semiconductor substrate on which rests an insulating layer having a semiconductor layer resting thereon, the method comprising a step of forming of at least two PN junctions formed at the interface between the substrate and the insulating layer.

[0026] According to an embodiment, the step of forming of the at least two PN junctions comprises the following steps:

[0027] a first masking step;

[0028] a first step of implantation of P-type dopants at the interface between the substrate and the insulating layer to form P-type doped wells;

[0029] a second masking step;

[0030] a second step of implantation of N-type dopants at the interface between the substrate and the insulating layer to form N-type doped wells; and

[0031] an anneal step.

[0032] According to an embodiment, the P-type dopants comprise boron.

[0033] According to an embodiment, the N-type dopants comprise arsenic.

[0034] According to an embodiment, the anneal step is a thermal ramp step performed by a laser.

[0035] According to an embodiment, the method comprises, after the step of forming of the at least two PN junctions, a step of manufacturing of the component.

[0036] According to an embodiment, the component is formed from the semiconductor layer.

BRIEF DESCRIPTION OF THE DRAWINGS

[0037] The foregoing features and advantages, as well as others, will be described in detail in the rest of the disclosure of specific embodiments given as an illustration and not limitation with reference to the accompanying drawings, in which:

[0038] FIG. 1 shows a cross-section view of an embodiment of an integrated circuit chip;

[0039] FIG. 2 illustrates a step of an implementation mode of a method of manufacturing the embodiment of FIG. 1;

[0040] FIG. 3 illustrates another step of an implementation mode of a method of manufacturing the embodiment of FIG. 1;

[0041] FIG. 4 illustrates another step of an implementation mode of a method of manufacturing the embodiment of FIG. 1;

[0042] FIG. 5 illustrates another step of an implementation mode of a method of manufacturing the embodiment of FIG. 1;

[0043] FIG. 6 illustrates another step of an implementation mode of a method of manufacturing the embodiment of FIG. 1;

[0044] FIG. 7 illustrates another step of an implementation mode of a method of manufacturing the embodiment of FIG. 1;

[0045] FIG. 8 illustrates another step of an implementation mode of a method of manufacturing the embodiment of FIG. 1;

[0046] FIG. 9 shows an embodiment of a chip comprising the embodiment of FIG. 1;

[0047] FIG. 10 shows an example of the chip of FIG. 9; and

[0048] FIG. 11 shows another example of the chip of FIG. 9.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

[0049] Like features have been designated by like references in the various figures. In particular, the structural and/or functional features that are common among the various embodiments may have the same references and may dispose identical structural, dimensional and material properties.

[0050] For clarity, only those steps and elements which are useful to the understanding of the described embodiments have been shown and are described in detail.

[0051] Unless indicated otherwise, when reference is made to two elements connected together, this signifies a direct connection without any intermediate elements other than conductors, and when reference is made to two elements coupled together, this signifies that these two elements can be connected or they can be coupled via one or more other elements.

[0052] In the following description, where reference is made to absolute position qualifiers, such as “front”, “back”, “top”, “bottom”, “left”, “right”, etc., or relative position qualifiers, such as “top”, “bottom”, “upper”, “lower”, etc., or orientation qualifiers, such as “horizontal”, “vertical”, etc., reference is made unless otherwise specified to the orientation of the drawings.

[0053] Unless specified otherwise, the expressions “about”, “approximately”, “substantially”, and “in the order of” signify plus or minus 10%, preferably of plus or minus 5%.

[0054] The embodiments described hereafter concern the electrical insulation of electronic chips, and more particularly the electrical insulation of integrated circuit chips adapted to receiving radio frequency signals, that is, AC electrical signals having a frequency in the range from 3 kHz to 30 GHz. The components in question herein are chips formed inside and/or on top of a structure of substrate-on-insulator type, or SOI-type structure. Leakage currents may happen to cross the insulating layer at the entrance of the SOI structure by capacitive effect, it is thus important to

correctly insulate such chips. A way enabling to overcome this problem is to form PN junctions at the interface between the buried insulating layer and the support substrate of the SOI-type structure. An electronic chip formed inside and/or on top of such a structure is described in relation with FIG. 1, and a method of manufacturing such a chip is described in relation with FIGS. 2 to 8.

[0055] Further, the embodiments described hereafter are particularly adapted to being used in integrated circuit chips used for radio frequency applications, that is, for applications using radio frequency signals. Examples of application of the embodiments are described in relation with FIGS. 9 to 11.

[0056] FIG. 1 is a cross-section view of an integrated circuit chip 100 formed inside and/or on top of an SOI-type structure 110 and comprising at least one electronic component 120.

[0057] The structure of SOI type 110 (Substrate On Insulator) comprises a stack comprising:

[0058] a semiconductor substrate 111 (P), also called support substrate 111;

[0059] a buried insulating layer 112 (BOX) resting on top of and in contact with an upper surface of support substrate 111; and

[0060] a semiconductor layer 113 resting on top of and in contact with an upper surface of buried insulating layer 112, opposite to a lower surface of insulating layer 112 in contact with the upper surface of support substrate 111.

[0061] Support substrate 111 is a substrate made of a semiconductor material, that is, of a material comprising one or a plurality of elements from column 14 of the periodic table of elements, such as silicon or germanium. According to a preferred embodiment, support substrate 111 is a high-resistivity substrate, that is, a substrate having a resistivity greater than or equal to 125 ohm-cm.

[0062] Buried layer 112 is an electrically-insulating layer having a thickness generally in the range from 10 to 400 nm, preferably from 200 to 400 nm. According to an embodiment, layer 112 is made of silicon oxide.

[0063] Semiconductor layer 113 is a layer made of a material comprising one or a plurality of elements from column 14 of the periodic table of elements, such as silicon or germanium. According to an example, semiconductor layer 113 has a thickness generally in the range from 40 nm to 500 μm, preferably in the range from 40 nm to 3 μm.

[0064] According to an embodiment, the SOI structure further comprises insulation enabling to prevent charge leakages from components formed based on semiconductor layer 113 into buried insulating layer 112 by capacitive effect. This insulation includes the forming of PN junctions at the interface between support substrate 111 and buried insulating layer 112. For this purpose, an alternation of P-type 114 (P) and N-type 115 (N) doped wells is formed at this interface. According to an embodiment, SOI structure 110 comprises at least two PN junctions formed at this interface, preferably more than 5 PN junctions, for example in the order of at least some ten PN junctions. In FIG. 1, in order not to overload the drawing, 6 PN junctions are shown. An implementation mode of a method of manufacturing such an SOI structure is described in detail in relation with FIGS. 2 to 8.

[0065] More specifically, there is here called alternation of N-type doped wells 114 and of P-type doped wells 115 a

succession of wells **114** and **115** in which each well **114** is surrounded by two wells **115**, and each well **115** is surrounded by two wells **114**. Thereby, PN junctions, as illustrated in FIG. 1, are formed along the interface between substrate **111** and buried insulating layer **112**.

[0066] According to an embodiment, wells **114** are portions of support substrate **111** extending from its upper surface, which are P-type doped. In other words, wells **114** have been doped by using one or a plurality of chemical elements forming part of column **13** of the periodic table of elements, such as boron. According to an example, wells **114** have a minimum thickness of 3 μm , and a minimum width of 200 nm. According to an example, well **114** comprises a concentration of dopant elements in the range from 10^{18} cm^{-3} to 10^{20} cm^{-3} , for example in the order of 10^{19} cm^{-3} .

[0067] According to an embodiment, wells **115** are portions of support substrate **111** extending from its upper surface which are N-type doped. In other words, wells **115** have been doped by using one or a plurality of chemical elements forming part of column **15** of the periodic table of elements, such as arsenic. According to an example, wells **115** have a minimum thickness of 3 μm , and a minimum width of 78 nm, for example in the range from 78 nm to 5 μm . According to an example, well **114** comprises a concentration of dopant elements in the range from 10^{18} cm^{-3} to 10^{20} cm^{-3} , for example in the order of 10^{19} cm^{-3} .

[0068] In the example of FIG. 1, electronic component **120** is a switch, such as a switch adapted to receiving radio frequency signals, that is, signals having a frequency in the range from 3 kHz to 30 GHz. More particularly, in the example of FIG. 1, component **120** is a metal-oxide-semiconductor field-effect transistor, or MOSFET transistor, or MOS transistor. In addition, component **120** is an N-channel MOS transistor, or N-type MOS transistor, or NMOS transistor.

[0069] Component **120**, or transistor **120**, is formed from semiconductor layer **113**. Transistor **120** comprises an N-doped source region **121** (N), formed across the entire thickness of a portion of semiconductor layer **113** (on the left-hand side in FIG. 1), and an N-doped drain region **122** (N), formed across the entire thickness of another portion of semiconductor layer **113** (on the right-hand side in FIG. 1). Regions **121** and **122** are separated by a P-doped channel region **123** (P). Channel region **123** is formed across the entire thickness of a portion of semiconductor layer **113** (at the center in FIG. 1).

[0070] Component **120** further comprises a gate stack comprising a gate insulator layer **124** formed on an upper surface of channel region **123**, and a gate layer **125** (N) resting on top of and contacting gate insulator layer **123**. Layers **124** and **125** have a width in the order of the width of channel region **123**.

[0071] Component **120** further comprises spacers **126** enabling to protect the lateral surfaces of the gate stack. Examples of spacers are illustrated in FIG. 1 and are within the abilities of those skilled in the art. In FIG. 1, spacers **126** comprise three insulating layers resting one on top of the other.

[0072] Component **120** further comprises silicide layers **127** arranged on an upper surface of source **121** and drain **122** regions and on an upper surface of gate layer **125**.

[0073] An advantage of the embodiment illustrated in FIG. 1 is that it enables to avoid charge leakages in buried insulating layer **112**.

[0074] FIGS. 2 to 9 are cross-section views illustrating steps of an implementation mode of a method of manufacturing a device of the type of the device **100** described in relation with FIG. 1.

[0075] At the step of FIG. 2, an SOI structure **200** of the type of the SOI structure **110** described in relation with FIG. 1 is considered. Structure **200** thus comprises:

[0076] a semiconductor support substrate **201** of the type of the support substrate **111** described in relation with FIG. 1;

[0077] a buried insulating layer **202** of the type of the buried insulating layer **112** described in relation with FIG. 1, layer **202** resting on top of and in contact with support substrate **202**; and

[0078] a semiconductor layer **203** of the type of the semiconductor layer **113** described in relation with FIG. 1, layer **203** resting on top of and in contact with buried insulating layer **202** on a surface opposite to support substrate **201**.

[0079] At the step of FIG. 3, a mask **204** is formed on an upper surface of semiconductor layer **203**, opposite to the surface of semiconductor layer **203** in contact with buried insulating layer **202**. Mask **204** enables to define the width of P-type doped wells of the type of the wells **114** described in relation with FIG. 1. According to an example, mask **204** is formed by a photolithography step.

[0080] At the step of FIG. 4, P-type doping elements are deeply implanted into the structure **200** of FIG. 3. More specifically, these doping elements form P-type doped wells **205** at the interface between buried insulating layer **202** and support substrate **201**. These wells **205** may extend over a portion of insulating layer **202** and a portion of support substrate **201**. According to an embodiment, the doping elements here are chemical elements forming part of column **13** of the periodic table of elements. According to an example, the doping elements comprise boron.

[0081] In the step of FIG. 5, mask **204** is removed from the upper surface of semiconductor layer **203**.

[0082] Further, at the step of FIG. 5, a mask **206** is formed on an upper surface of semiconductor layer **203**, opposite to the surface of semiconductor layer **203** in contact with buried insulating layer **202**. Mask **206** enables to define the width of N-type doped wells of the type of the wells **115** described in relation with FIG. 1. According to an example, mask **206** is formed by a photolithography step.

[0083] At the step of FIG. 6, N-type dopant elements are deeply implanted into the structure **200** of FIG. 5. More specifically, these doping elements form N-type doped wells **207** at the interface between buried insulating layer **202** and support substrate **201**, designated in FIG. 6 by a dotted line **208**. These wells **207** may extend over a portion of insulating layer **202** and a portion of support substrate **201**. According to an embodiment, the doping elements here are chemical elements forming part of column **15** of the periodic table of elements. According to an example, the doping elements comprise arsenic.

[0084] According to an embodiment, and as shown in relation with FIG. 1, the P-type **205** and N-type **207** doped wells are arranged alternately. More specifically, each well **204** is surrounded by two wells **205**, and each well **205** is surrounded by two wells **204**. Thereby, PN junctions, as illustrated in FIG. 6, are formed along the interface between substrate **201** and buried insulating layer **202**. To obtain such

a result, masks **204** and **206** have periodic openings in opposition with respect to each other.

[0085] At the step of FIG. 7, an annihilation operation is implemented. This step enables to make layer **202** insulating again. According to an example, the annihilation operation is a heat treatment step, for example an anneal step or a thermal ramp step. According to a specific example, the annihilation operation is a thermal ramp step carried out by a laser, with a temperature varying between 500 and 600° C.

[0086] The step of FIG. 7 enables to migrate the doping elements present in buried insulating layer **202** and, more generally, to decrease the thickness of wells **205** and **207**.

[0087] At the step of FIG. 8, a component **209**, of the type of component **120**, is formed from semiconductor layer **203**. The steps of manufacturing such a component are within the abilities of those skilled in the art.

[0088] According to an embodiment, the dimensions of wells **205** and **207** are established according to the different stages of manufacturing of component **209**. Indeed, the steps of manufacturing of component **209** may comprise heat treatment operations capable of having the doping elements of the wells **205** and **207** diffuse, and this phenomenon is to be avoided. This is why wells **207** have a minimum thickness of 3 μm, and a minimum width of 200 nm, and wells **207** have a minimum thickness of 3 μm, and a minimum width of 78 nm. In practice, wells **207** have a width much greater than 78 nm, for example in the order of several hundreds of nm.

[0089] FIG. 9 shows an example of an integrated circuit chip **901** (RF) of the type of the integrated circuit chip **100** described in relation with FIG. 1.

[0090] As previously mentioned, an application of the chip **100** described in relation with FIG. 1 may be an integrated circuit chip formed on an SOI structure of the type of the structure **110** described in relation with FIG. 1, an earlier version of which comprises problems of current leakage in the buried insulating layer of the structure.

[0091] More particularly, integrated circuit chip **901** represents a preferred application of the chip **100** of FIG. 1, since chip **901** is adapted to using radio frequency signals. More precisely, chip **901** comprises at least one electronic component **902** (HW) of the type of the component **120** described in relation with FIG. 1, adapted to using one or a plurality of radio frequency signals. As previously described, component **902** may be any electronic component adapted to processing one or a plurality of radio frequency signals, such as a switch, a transmission line, or an integrated inductor.

[0092] FIG. 10 shows another example of an integrated circuit chip **1000** of the type of the integrated circuit chip **100** described in relation with FIG. 1.

[0093] Integrated circuit chip **1000** is a switch or relay of SPDT (Single Pole Double Throw) type intended to be controlled by radio frequency signals, for example signals using home WiFi communication protocols.

[0094] According to an example, integrated circuit chip **1000** comprises an input node **RX1000** and an output node **TX1000** adapted, respectively, to receiving and to transmitting radio frequency signals.

[0095] Integrated circuit chip **1000** further comprises a connection to an antenna **ANT1000**, and a connection **PDET100** enabling to collect an image of the output power

of the antenna. This image is obtained by using a signal sampled from a directional coupler **INV1001** described hereafter.

[0096] Integrated circuit chip **1000** comprises, on a first branch of reception of a signal by the antenna, a capacitor **C1001**, an amplifier **LNA1000**, a filtering circuit **F1001**, and a switch **SW1001**. A first conduction terminal of capacitor **C1001** is coupled, preferably connected, to input node **RX1000**, and a second conduction terminal of capacitor **C1001** is coupled, preferably connected, to an output of amplifier **LNA1000**. An input of amplifier **LNA1000** is coupled, preferably connected, to an output of filtering circuit **F1001**. According to an example, amplifier **LNA1000** is a low-noise amplifier. Switch **SW1001** is arranged in parallel with amplifier **LNA1000** and with filtering circuit **F1001**. More particularly, a conduction terminal of switch **SW1001** is coupled, preferably connected, to the output of amplifier **LNA1000**, and a second conduction terminal of switch **SW1001** is coupled, preferably connected, to an input of filtering circuit **F1001**.

[0097] Integrated circuit chip **1000** further comprises a switch **SW1002**, having an input coupled, preferably connected, to antenna **ANT1000** and comprising two outputs. A first output is coupled to the first branch and, more precisely, is coupled, preferably connected, to the input of filtering circuit **F1001**. A second output is coupled to a second transmission branch of the chip **1000** described hereafter.

[0098] Integrated circuit chip **1000** further comprises a coil **B1001** coupling antenna **ANT1000** to a terminal receiving a reference potential, for example the ground. Thus, a first terminal of coil **B1001** is coupled, preferably connected, to antenna **ANT1001**, and a second terminal of coil **B1001** is coupled, preferably connected, to the node receiving the reference potential.

[0099] Integrated circuit chip **1000** further comprises a second branch for transmitting signals to antenna **ANT1000**, comprising a capacitor **C1002**, an amplifier **PA1001**, a filtering circuit **F1002**, directional coupler **INV1001**, and a diode **D1001**. A first conduction terminal of capacitor **C1002** is coupled, preferably connected, to input node **TX1000**, and a second conduction terminal of capacitor **C1002** is coupled, preferably connected, to an input of amplifier **PA1001**. An output of amplifier **PA1001** is coupled, preferably connected, to an output of filter circuit **F1002**. According to an example, amplifier **PA1001** is a controllable-gain amplifier. An input of inverter **IV1001** is coupled, preferably connected, to the output of filter circuit **F1002**. A first output of coupler **INV1001** is coupled, preferably connected, to the second output of switch **SW1002**, and a second output of coupler **INV1001** is coupled, preferably connected, to an anode of diode **D1001**. Directional coupler **INV1001** enables to sample a fraction of the signal arriving at the antenna, to be able to determine an image of the power level emitted by this antenna. The cathode of diode **D1001** is coupled, preferably connected, to connection **PDET100**.

[0100] According to an example, switch **SW1002** may be the component **120** described in relation with FIG. 1.

[0101] FIG. 11 shows another example of an integrated circuit chip **1100** of the type of the integrated circuit chip **100** described in relation with FIG. 1.

[0102] Integrated circuit chip **1100** is a radio frequency switch with a frequency band selection.

[0103] Integrated circuit chip 1100 comprises a control interface INT1100 (MIPI Interface) and a set of switches SW1100.

[0104] Interface INT1100 is adapted to controlling the set of switches SW1100 and receives, for this purpose, a communication signal VIO1100, a clock signal CLK1100, and a data signal SDATA1100. Interface INT1100 is further coupled to a reference potential, for example ground GND1100.

[0105] The set of switches SW1100 comprises N switches, N being an integer greater than one, all having a common input coupled to an antenna ANT1100, and distinct outputs. More particularly, each switch of set SW1100 has a first conduction terminal coupled, preferably connected, to antenna ANT1100, and a second conduction terminal coupled, preferably connected, to an output of set SW1100. In the example illustrated in FIG. 11, the set comprises eight (8) switches.

[0106] According to an example, the set of switches SW1100 may be the component 120 described in relation with FIG. 1.

[0107] Various embodiments and variants have been described. Those skilled in the art will understand that certain features of these various embodiments and variants may be combined, and other variants will occur to those skilled in the art.

[0108] Finally, the practical implementation of the described embodiments and variants is within the abilities of those skilled in the art based on the functional indications given hereabove.

What is claimed:

1. An integrated circuit chip comprising:
 - a silicon-on-insulator structure comprising:
 - a semiconductor substrate;
 - an insulating layer disposed on the semiconductor substrate; and
 - a semiconductor layer disposed on the insulating layer;
 - at least two PN junctions disposed at an interface between the semiconductor substrate and the insulating layer; and
 - at least one component, arranged inside and/or on top of the silicon-on-insulator structure.
2. The integrated circuit chip according to claim 1, wherein the at least two PN junctions are disposed against the interface between the semiconductor substrate and the insulating layer.
3. The integrated circuit chip according to claim 1, wherein the at least two PN junctions comprise an alternation of N-type doped wells and of P-type doped wells.
4. The integrated circuit chip according to claim 3, wherein the N-type doped wells have a width greater than or equal to 78 nm.
5. The integrated circuit chip according to claim 3, wherein the P-type doped wells have a width greater than or equal to 200 nm.
6. The integrated circuit chip according to claim 1, wherein the semiconductor substrate has a resistivity greater than 125 ohm-cm.
7. The integrated circuit chip according to claim 1, wherein the at least one component is at least one switch.

8. The integrated circuit chip according to claim 7, wherein the at least one component is at least one metal-oxide-semiconductor (MOS)-type transistor.

9. The integrated circuit chip according to claim 1, wherein the at least one component is configured to use alternating current (AC) electrical signals at frequencies in a range from 3 kHz to 30 GHz.

10. The integrated circuit chip according to claim 9, wherein the integrated circuit chip is a single-pole double-throw (SPDT)-type switch, and the at least one component is at least one switch.

11. The integrated circuit chip according to claim 9, wherein the integrated circuit chip is a radio frequency switch with a frequency band selection, and the at least one component is at least one switch.

12. The integrated circuit chip according to claim 1, wherein the at least two PN junctions comprise more than five PN junctions.

13. A method of manufacturing an integrated circuit chip, the method comprising:

providing a silicon-on-insulator structure comprising:

a semiconductor substrate;

an insulating layer disposed on the semiconductor substrate; and

a semiconductor layer disposed on the insulating layer;

forming at least two PN junctions at an interface between the semiconductor substrate and the insulating layer; and

forming at least one component inside and/or on top of the silicon-on-insulator structure.

14. The method according to claim 13, wherein the forming the at least two PN junctions comprises:

forming a first mask on the silicon-on-insulator structure;

implanting P-type dopants at the interface between the semiconductor substrate and the insulating layer to form P-type doped wells;

forming a second mask on the silicon-on-insulator structure;

implanting N-type dopants at the interface between the semiconductor substrate and the insulating layer to form N-type doped wells; and

annealing the integrated circuit chip.

15. The method according to claim 14, wherein the P-type dopants comprise boron.

16. The method according to claim 14, wherein the N-type dopants comprise arsenic.

17. The method according to claim 14, wherein the annealing comprises a thermal ramp using by a laser.

18. The method according to claim 13, further comprising forming the at least one component after forming the at least two PN junctions.

19. The method according to claim 18, further comprising forming the at least one component at least in part using the semiconductor layer.

20. The method according to claim 13, wherein forming the at least two PN junctions comprises forming more than five PN junctions.

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